

SARYKOL, Kazakhstan

WMO: 288670

Lat: 53.327N

Lon: 65.524E

Elev: 709

StdP: 14.32

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.6	-18.9	-30.8	1.0	-24.4	-25.0	1.4	-18.8	33.7	13.4	27.7	12.6	4.6	20	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	88.7	67.1	85.4	66.2	82.2	65.0	69.5	84.7	67.7	82.4	66.1	80.1	8.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	90.6	79.2	61.7	84.5	77.0	59.9	79.1	75.2	34.0	84.5	32.5	82.4	31.2	80.1	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	27.8	23.2	20.2	DB	-28.6	94.2	6.6	3.4	-33.4	96.6	-37.3	98.6	-41.0	100.5	-45.8	103.0
(5)				WB	-29.0	72.6	6.1	2.7	-33.4	74.5	-37.0	76.0	-40.4	77.5	-44.9	79.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	2.4	4.4	19.5	41.6	56.8	65.8	67.6	65.2	54.3	40.4	21.9	8.4	
	DBStd	25.94	12.16	11.65	9.95	10.86	8.93	7.63	6.48	7.68	8.68	8.91	12.64	12.43	
	HDD50	6531	1475	1276	944	293	37	2	0	1	49	320	844	1290	
	HDD65	10405	1940	1696	1409	703	281	81	48	96	338	764	1294	1755	
	CDD50	1980	0	0	0	40	247	478	545	471	178	21	0	0	
	CDD65	378	0	0	0	0	26	106	128	101	17	0	0	0	
	CDH74	3783	0	0	0	13	333	1119	1179	945	193	1	0	0	
	CDH80	1167	0	0	0	1	74	371	352	310	59	0	0	0	
Wind	WSAvg	7.5	7.6	8.3	8.1	7.7	8.6	7.6	6.8	6.3	6.8	7.1	7.5	8.0	
Precipitation	PrecAvg	13.9	0.7	0.6	0.7	0.9	1.4	1.7	2.2	1.7	1.0	1.2	1.1	0.8	
	PrecMax	19.6	2.0	2.1	2.1	2.6	4.1	3.9	5.7	4.1	2.2	2.5	2.3	1.9	
	PrecMin	7.2	0.1	0.0	0.0	0.0	0.1	0.2	0.7	0.2	0.1	0.1	0.2	0.2	
	PrecStd	3.1	0.5	0.4	0.5	0.6	0.9	1.1	1.3	1.2	0.6	0.7	0.6	0.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.7	33.2	45.9	75.2	86.3	93.0	92.5	91.8	87.4	70.4	48.1	34.0	
		MCWB	30.6	30.6	40.9	57.6	64.4	67.9	69.2	68.4	64.8	55.4	42.8	31.0	
	2%	DB	27.7	29.7	37.9	69.0	81.6	88.1	87.3	87.3	79.2	64.3	43.0	31.1	
		MCWB	25.9	27.8	34.4	55.5	63.4	65.7	68.5	67.3	61.3	51.9	39.3	29.1	
	5%	DB	24.7	25.4	35.0	63.7	77.5	84.3	83.8	83.3	73.6	59.2	39.7	28.4	
		MCWB	23.1	23.8	32.0	52.8	61.4	65.2	66.6	66.1	59.0	49.0	36.6	26.6	
	10%	DB	21.2	21.3	33.1	58.7	73.3	80.4	80.5	79.3	68.8	54.1	36.2	25.2	
		MCWB	19.9	19.7	30.5	49.8	59.0	63.5	65.3	64.4	57.0	46.3	33.4	23.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.6	31.0	40.9	60.3	68.1	70.8	72.0	70.9	66.4	56.4	43.7	31.5
			MCDB	31.8	32.9	45.4	72.1	83.0	87.6	86.9	87.0	84.6	68.6	47.1	33.4
2%		WB	26.0	27.9	34.7	56.2	64.7	68.1	69.9	68.7	62.6	52.5	39.5	29.1	
		MCDB	27.4	29.8	37.3	68.2	78.8	84.3	84.5	84.3	76.0	62.7	42.9	31.1	
5%		WB	23.2	23.8	32.3	52.9	62.1	66.3	67.9	66.8	59.6	49.4	36.7	26.6	
		MCDB	24.6	25.5	34.8	63.2	76.0	81.8	81.3	81.2	72.1	58.3	39.6	28.4	
10%		WB	19.9	19.7	30.4	50.0	59.3	64.3	66.2	64.8	57.1	46.5	33.6	23.5	
		MCDB	21.3	21.3	33.0	58.2	72.2	78.6	79.1	78.0	68.3	53.6	35.9	25.1	
Mean Daily Temperature Range	5% DB	MDBR	12.7	13.6	14.1	17.1	21.1	20.5	18.7	19.4	19.3	15.4	11.2	12.7	
		MCDBR	12.7	14.1	12.8	25.8	26.7	25.6	23.8	25.4	26.3	24.3	12.8	12.7	
		MCWBR	12.2	13.2	11.1	16.8	15.1	12.7	11.8	12.7	13.9	15.2	10.8	11.8	
	5% WB	MCDBR	12.5	13.7	12.6	24.3	25.0	23.7	21.6	23.5	24.6	22.6	12.4	12.7	
		MCWBR	12.2	12.9	11.1	16.7	15.8	13.8	12.3	13.2	14.4	14.9	10.8	12.0	
Clear-Sky Solar Irradiance	taub	0.288	0.321	0.372	0.408	0.407	0.403	0.418	0.427	0.374	0.352	0.320	0.288		
	taud	2.065	2.016	1.971	2.132	2.222	2.308	2.263	2.209	2.308	2.264	2.157	2.050		
	Ebn at Noon	206	237	249	259	267	269	262	251	251	227	193	178		
	Edh at Noon	23	34	44	43	42	39	40	40	32	27	22	20		
All-Sky Solar Radiation	RadAvg	360	721	1195	1491	1830	1968	1839	1523	1104	586	285	238		
	RadStd	36	34	111	157	155	146	142	149	83	81	31	37		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	+1.13	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page